Business Question and Visualization Report

Date	28 October 2026
Team ID	EN25CS3010457
Project Name	EV Market Performance and Insights
Maximum Marks	5 Marks

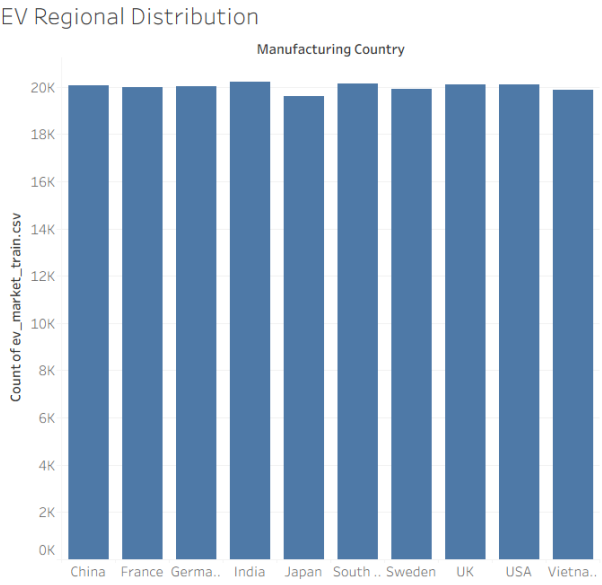
Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. Which countries and vehicle categories lead EV production and annual sales?

- *Visualization*: Stacked bar chart of annual sales units by country and vehicle category.
- *Screenshot of visualisation* – EV Regional Distribution



- *Description of visualisation*

  Visualization Overview: This column chart shows the count of records in the ev_market_train.csv dataset for each manufacturing country: China, France, Germany, India, Japan, South Korea, Sweden, UK, USA and Vietnam. It describes how the EV records are spread across producing countries. It is a record count, not a sales or production volume.

  Key Observations: All ten columns are almost the same height, at about 20K records each. India is marginally the highest (about 20.2K) and Japan is marginally the lowest (about 19.6K). The gap between the highest and lowest country is very small, and no country clearly stands out.

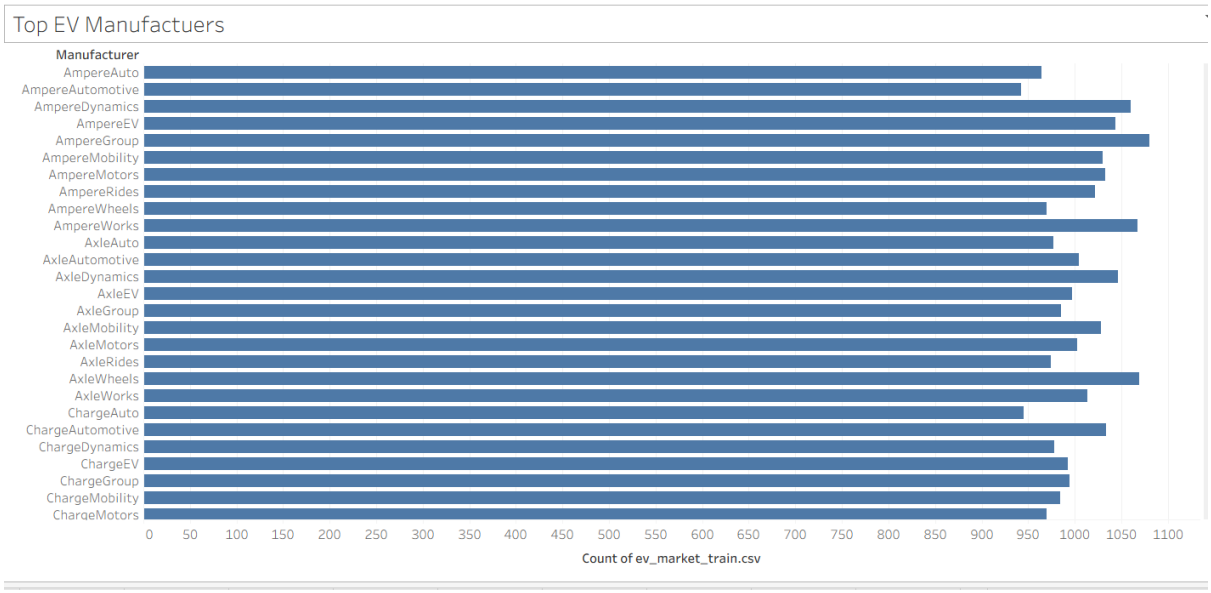
  Business Insights: For the EV Market Performance and Insights project, this means the dataset covers all ten countries evenly, so comparisons between countries are not biased by unequal data coverage. Because the record counts are so similar, they cannot show which country leads in production. That ranking needs annual sales units by country.

  Conclusion: EV records are evenly distributed across the ten manufacturing countries, so the dataset gives balanced country coverage and country leadership must be judged on sales units rather than record counts.

2. Which manufacturers perform best on customer reviews and sales units?

- *Visualization*: Packed bubble chart of manufacturers by customer review and sales units.

- *Screenshot of visualisation* – Top EV Manufacturers



- *Description of visualisation*

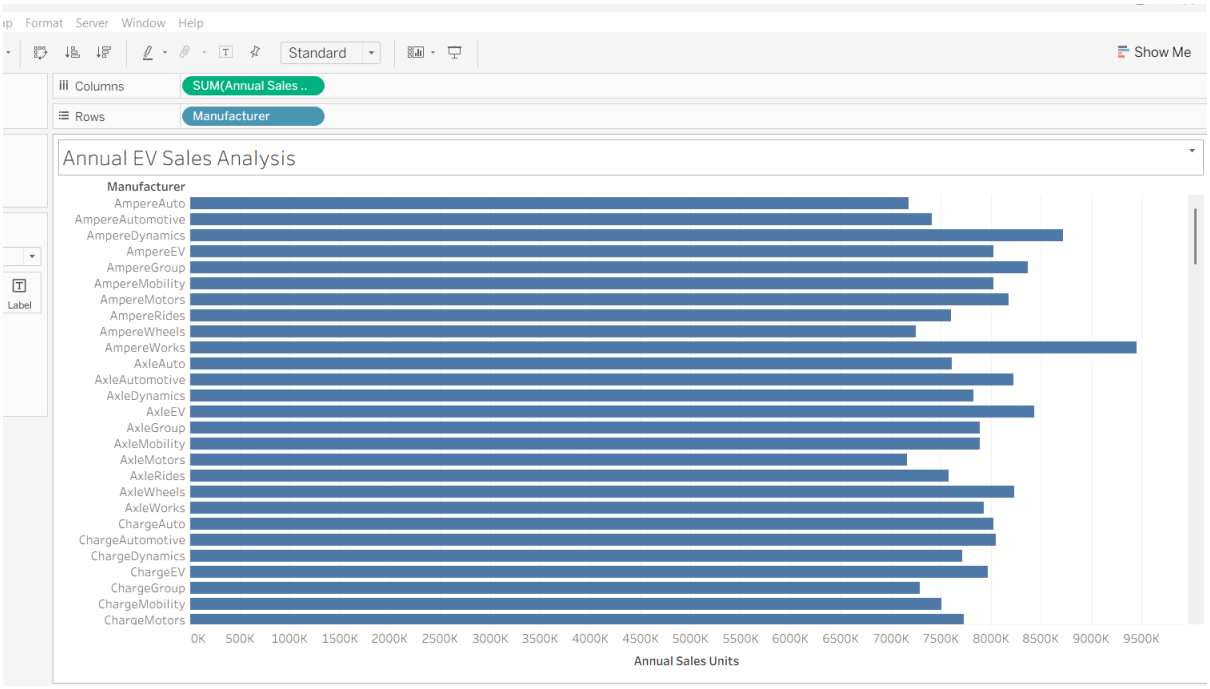
  Visualization Overview: This horizontal bar chart shows the count of records in ev_market_train.csv for each manufacturer. The screenshot displays the Ampere, Axle and Charge manufacturer groups (the list continues beyond the visible rows).

  Key Observations: The counts of the visible manufacturers range from about 940 to about 1,080. AmpereGroup has the highest count (about 1,080), followed closely by AxleWheels (about 1,068), AmpereWorks (about 1,066) and AmpereDynamics (about 1,060). AmpereAutomotive (about 940) and ChargeAuto (about 945) have the lowest counts. Most bars lie between 960 and 1,050.

  Business Insights: The manufacturers have similar numbers of models/records in the dataset, so none of them dominates by market presence alone. A larger count only means more entries in the data, so it should be combined with sales units and customer ratings before calling a manufacturer a top performer.

  Conclusion: Manufacturer record counts are close to each other, with AmpereGroup slightly ahead; presence in the dataset alone does not decide the best manufacturer.

- *Screenshot of visualisation* – Annual EV Sales Analysis



- *Description of visualisation*

  Visualization Overview: This horizontal bar chart shows the sum of annual sales units for each manufacturer (Tableau shelf: SUM of Annual Sales on Columns, Manufacturer on Rows). The axis runs from 0K to above 9,500K units, and the visible rows cover the Ampere, Axle and Charge manufacturers.

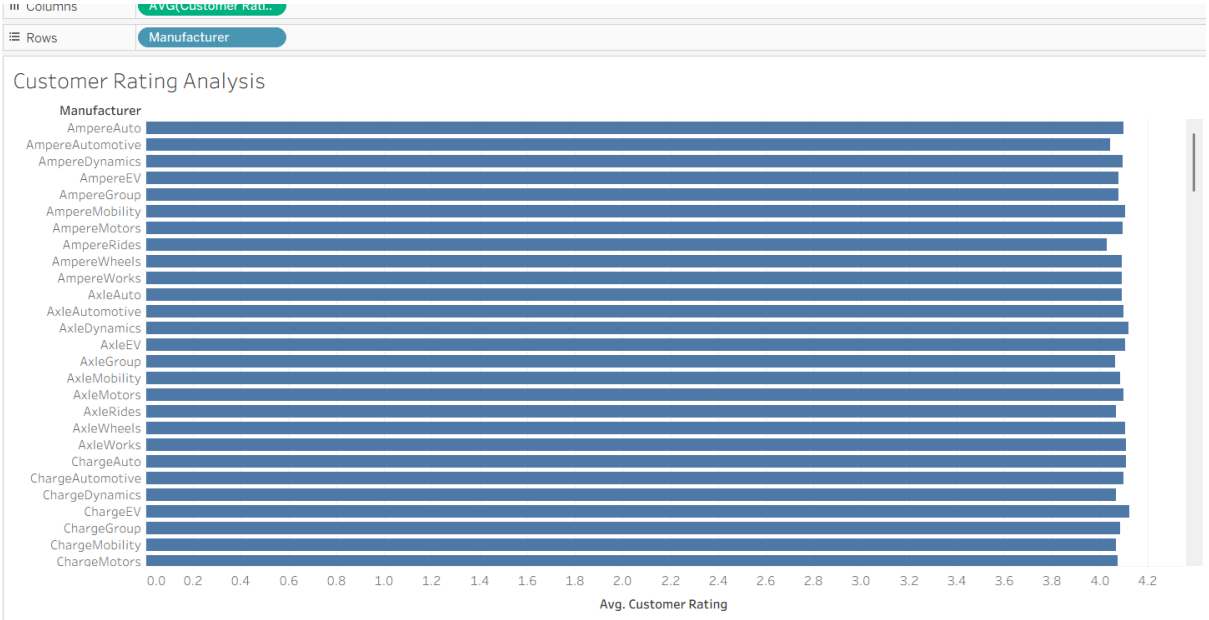
  Key Observations: AmpereWorks has the highest sales, at about 9.4 million units. AmpereDynamics is second among the visible rows (about 8.7 million), followed by AxleEV and AmpereGroup (about 8.3 to 8.4 million). The lowest bars are AxleMotors,

AmpereAuto and AmpereWheels, at roughly 7.1 to 7.2 million units. Overall, sales vary from about 7.1 million to 9.4 million units.

Business Insights: Unlike the record counts, annual sales show a clear difference between manufacturers: the top seller sells roughly 2 million units more than the lowest. This is the key measure for identifying the sales leaders, and it points to AmpereWorks and AmpereDynamics as the strongest performers in the visible group.

Conclusion: Sales units separate the manufacturers much more clearly than record counts do, with AmpereWorks leading the visible manufacturers.

- *Screenshot of visualisation* – Customer Rating Analysis



- *Description of visualisation*

Visualization Overview: This horizontal bar chart shows the average customer rating (AVG of Customer Rating) for each manufacturer, on a scale from 0.0 to about 4.2.

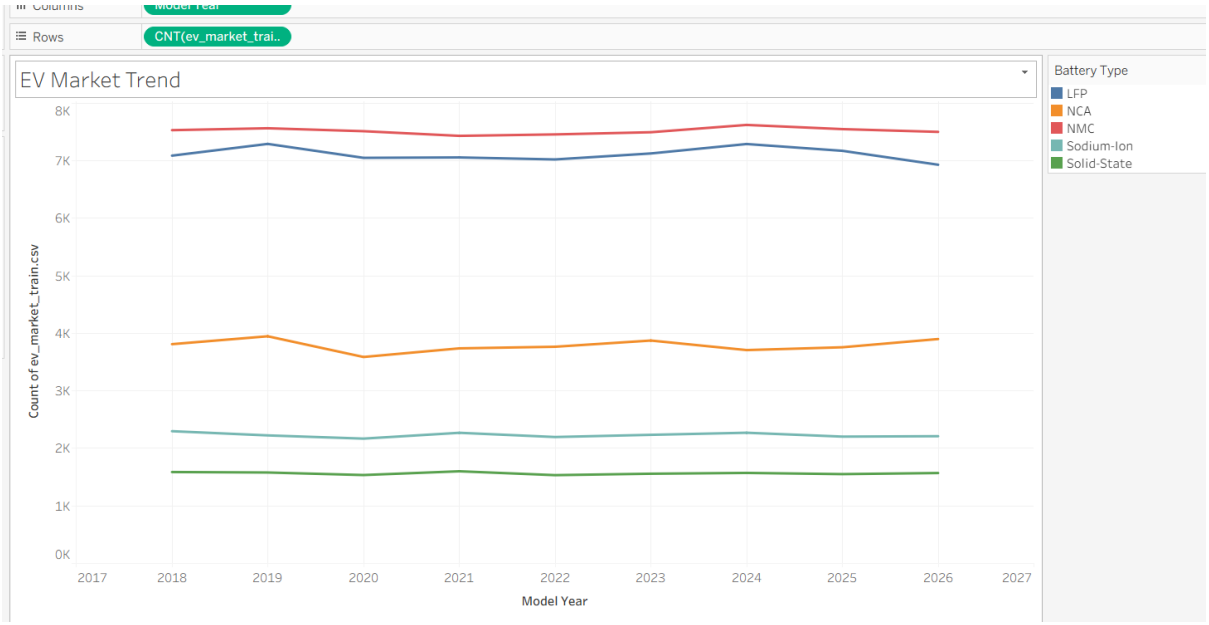
Key Observations: All visible manufacturers have very similar ratings, at roughly 4.0 to 4.1. ChargeEV and AxleDynamics are the highest (about 4.12), while AmpereRides (about 4.03) and AmpereAutomotive (about 4.04) are the lowest. The difference between the best and the lowest rated manufacturer is only about 0.1.

Business Insights: Customer satisfaction is uniformly high across the manufacturers, so ratings alone do not separate the best performers. When compared with the sales chart, manufacturers with very different sales (for example AmpereWorks, with the highest sales, and AxleMotors, with one of the lowest) have almost the same rating, so sales differences are not explained by customer reviews in this data.

Conclusion: Customer ratings are consistently around 4.1 for all manufacturers, so sales units are the stronger indicator of which manufacturers perform best.

3. How have annual sales trended over the years for each category?

- *Visualization*: Multi-line chart of annual sales units by model year.
- *Screenshot of visualisation* – EV Market Trend



- *Description of visualisation*

Visualization Overview: This multi-line chart shows the count of records (CNT of ev_market_train.csv) by Model Year, from 2018 to 2026, with one line for each battery type: LFP, NCA, NMC, Sodium-Ion and Solid-State. It is shown by battery type, not by vehicle category.

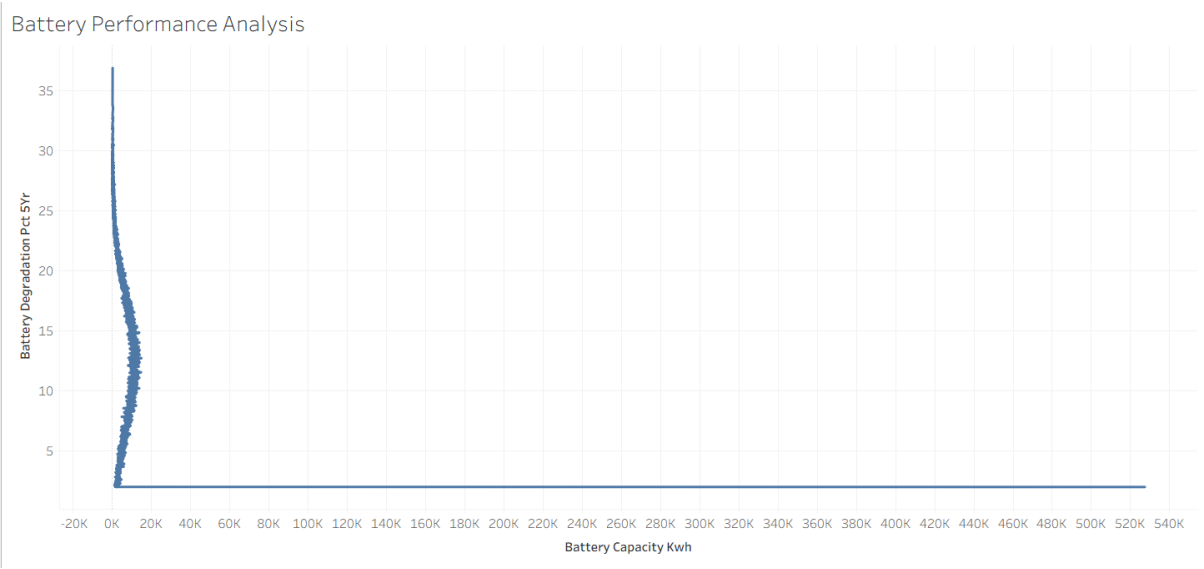
Key Observations: All five lines are almost flat across the years. NMC is the highest line at about 7.4K to 7.7K (peak around 2024), LFP follows at about 6.9K to 7.3K (lowest point in 2026), NCA stays near 3.6K to 3.9K, Sodium-Ion near 2.2K to 2.3K and Solid-State lowest at about 1.5K to 1.6K. The order of the battery types does not change in any year.

Business Insights: The market mix of battery technologies is stable over the model years shown, with NMC and LFP clearly the most common. No battery type shows strong growth or decline, and Solid-State and Sodium-Ion remain smaller segments. Manufacturers and planners can therefore expect NMC and LFP to remain the main technologies in this dataset.

Conclusion: There is no major year-to-year change; the trend is steady, with NMC and LFP leading in every model year.

4. Which battery technologies dominate sales, and how do they compare on cycle life?

- *Visualization*: Dual-axis bar and line chart of sales units and average cycle life by battery type.
- *Screenshot of visualisation* – Battery Performance Analysis



- *Description of visualisation*

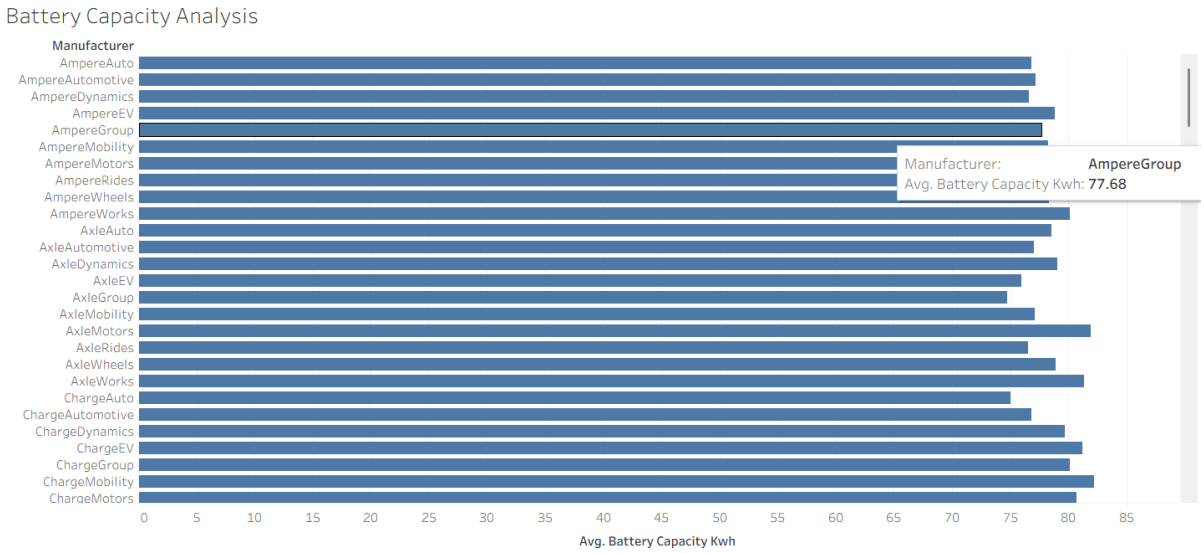
Visualization Overview: This line chart plots Battery Capacity (kWh) on the horizontal axis against Battery Degradation Pct 5Yr (battery degradation after five years, in percent) on the vertical axis.

Key Observations: Degradation values range from about 2% to about 37%. Most of the plotted values lie close to the vertical axis (below about 15K on the capacity axis), with the widest spread at around 10% to 17% degradation. Higher degradation values (above 25%) have very small values on the capacity axis. A long flat line at about 2% degradation extends to about 527K on the capacity axis.

Business Insights: The capacity axis shows values in thousands, far higher than a single battery (the average capacity chart below shows about 75 to 82 kWh). This indicates that capacity is being summed over many records, so the chart shows totals rather than individual vehicles. It also shows that the largest share of capacity is linked to the lowest degradation level (about 2%), and that very high degradation (above 25%) is uncommon. For a clearer comparison, an average capacity or scatter view is recommended.

Conclusion: Most batteries show low to moderate degradation after five years, and very high degradation is rare, although the chart would be easier to interpret with averaged values.

- *Screenshot of visualisation* – Battery Capacity Analysis



- *Description of visualisation*

Visualization Overview: This horizontal bar chart shows the average battery capacity (kWh) for each manufacturer. A tooltip in the screenshot confirms the value for AmpereGroup: 77.68 kWh.

Key Observations: The average capacity of the visible manufacturers lies between about 75 and 82 kWh. ChargeMobility (about 82 kWh), AxleMotors (about 82 kWh) and AxleWorks (about 81 kWh) have the largest batteries, while AxleGroup (about 75 kWh) and ChargeAuto (about 75 kWh) have the smallest.

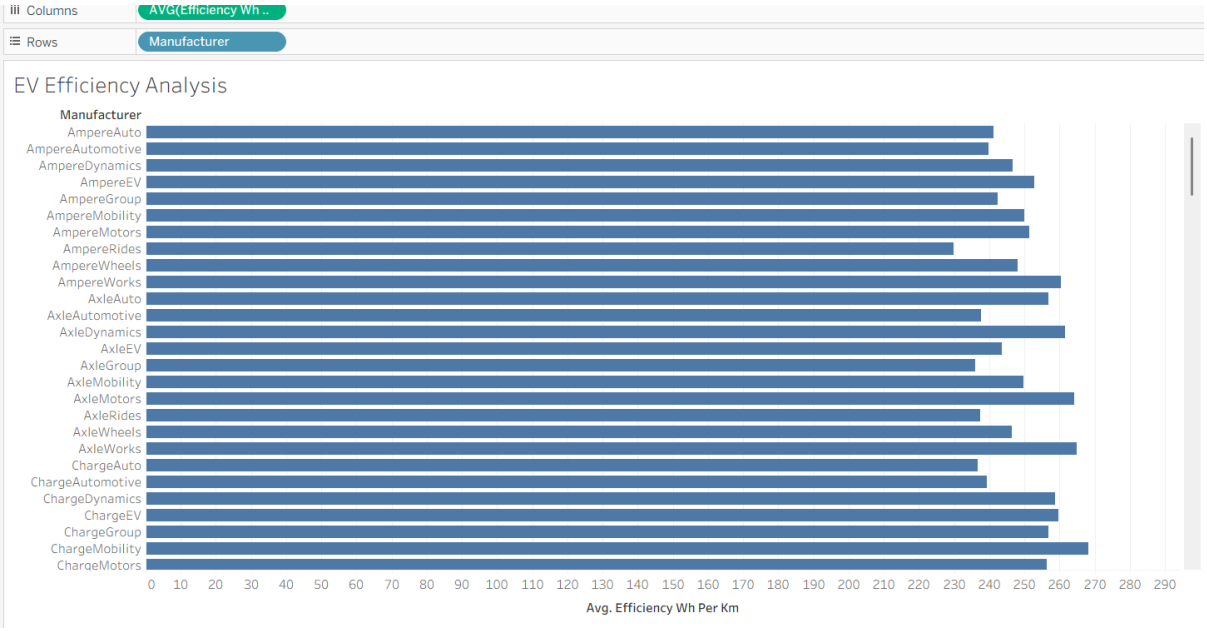
Business Insights: Battery size is quite similar across manufacturers, with only about a 7 kWh gap between the highest and the lowest. Manufacturers with larger batteries are positioned for longer range, but, as the charging and efficiency charts in Question 5 show, larger batteries also go with longer charging times and higher energy use per km.

Conclusion: Average battery capacity is consistently high (about 75 to 82 kWh), and the differences between manufacturers are small.

5. How do charging speed and energy efficiency relate across vehicles?

- *Visualization*: Bubble (scatter) chart of max DC charging kW vs efficiency Wh per km.

- *Screenshot of visualisation* – EV Efficiency Analysis



- *Description of visualisation*

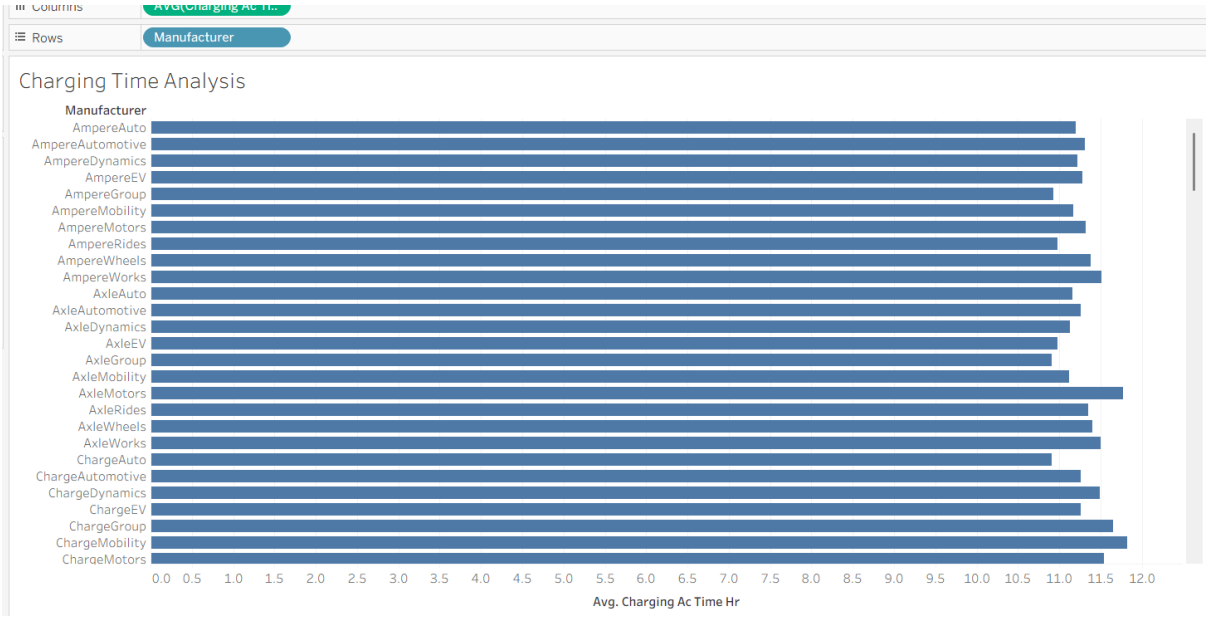
Visualization Overview: This horizontal bar chart shows the average energy efficiency (AVG of Efficiency Wh per Km) for each manufacturer. A lower Wh/km value means the vehicle uses less energy per kilometre.

Key Observations: The visible manufacturers range from about 230 Wh/km to about 268 Wh/km. AmpereRides (about 230) is the most efficient. AxleGroup, ChargeAuto and AmpereAutomotive are also low (about 236 to 240). ChargeMobility (about 268), AxleWorks (about 265) and AxleMotors (about 264) use the most energy per km.

Business Insights: The gap of roughly 38 Wh/km between the best and the least efficient manufacturer affects running cost and real driving range. The manufacturers with the highest consumption (ChargeMobility, AxleMotors) also have the largest average batteries (Question 4), which suggests that bigger batteries come with higher energy use per km.

Conclusion: Efficiency differs noticeably between manufacturers (about 230 to 268 Wh/km), and AmpereRides is the most efficient of those visible.

- *Screenshot of visualisation* – Charging Time Analysis



- *Description of visualisation*

Visualization Overview: This horizontal bar chart shows the average AC charging time in hours (AVG of Charging AC Time Hr) for each manufacturer. It uses AC charging time as the charging measure.

Key Observations: Average charging time is long for all visible manufacturers, between about 10.9 and 11.8 hours. ChargeMobility (about 11.8 hours) and AxleMotors (about 11.8 hours) take the longest, while AxleGroup, ChargeAuto and AmpereGroup (about 10.9 hours) and AmpereRides (about 11 hours) take the shortest.

Business Insights: The manufacturers with the longest charging times (ChargeMobility and AxleMotors) are also those with the highest energy use per km and the largest batteries, and those with the shortest times (AxleGroup, ChargeAuto) have the smaller batteries and lower energy use. This points to a consistent link between battery size, efficiency and charging time. Faster charging would be an advantage for manufacturers, since about 11 hours on AC is long.

Conclusion: Charging time moves with battery size and efficiency; all manufacturers need around 11 to 12 hours for an AC charge, with ChargeMobility and AxleMotors the slowest.